

## ID: 24101230

So the packet sizes are:

1. 2410
2. 1230
3. 142 (reverse of 2410)
4. 321 (reverse of 1230)

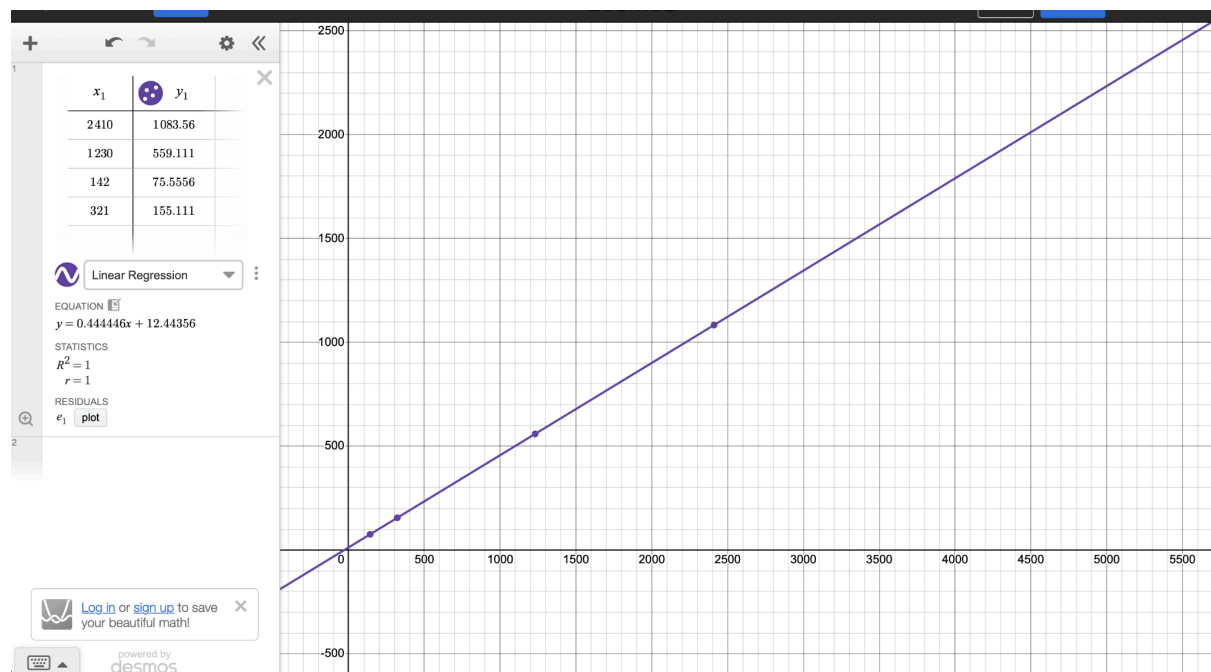
```
TraceDelay TX 2410 bytes to 10.1.1.2 Uid: 0 Time: +2s
TraceDelay: RX 2410 bytes from 10.1.1.1 Sequence Number: 0 Uid: 0 TXtime: +2e+09ns RXtime: +2.00594e+09ns Delay: +5.9392e+06ns
FlowID: 1 (UDP 10.1.1.1/49153-->10.1.1.2/9)
Tx Bytes: 2438
Rx Bytes: 2438
Mean Delay: 0.0059392s
Throughput: 1083.56bps
```

```
TraceDelay TX 1230 bytes to 10.1.1.2 Uid: 0 Time: +2s
TraceDelay: RX 1230 bytes from 10.1.1.1 Sequence Number: 0 Uid: 0 TXtime: +2e+09ns RXtime: +2.00402e+09ns Delay: +4.016e+06ns
FlowID: 1 (UDP 10.1.1.1/49153-->10.1.1.2/9)
Tx Bytes: 1258
Rx Bytes: 1258
Mean Delay: 0.004016s
Throughput: 559.111bps
prantik@Prantiks-MacBook-Air ns-3.43 %
```

```
TraceDelay TX 142 bytes to 10.1.1.2 Uid: 0 Time: +2s
TraceDelay: RX 142 bytes from 10.1.1.1 Sequence Number: 0 Uid: 0 TXtime: +2e+09ns RXtime: +2.00228e+09ns Delay: +2.2752e+06ns
FlowID: 1 (UDP 10.1.1.1/49153-->10.1.1.2/9)
Tx Bytes: 170
Rx Bytes: 170
Mean Delay: 0.0022752s
Throughput: 75.5556bps
```

```
TraceDelay TX 321 bytes to 10.1.1.2 Uid: 0 Time: +2s
TraceDelay: RX 321 bytes from 10.1.1.1 Sequence Number: 0 Uid: 0 TXtime: +2e+09ns RXtime: +2.00256e+09ns Delay: +2.5616e+06ns
FlowID: 1 (UDP 10.1.1.1/49153-->10.1.1.2/9)
Tx Bytes: 349
Rx Bytes: 349
Mean Delay: 0.0025616s
Throughput: 155.111bps
```

## GRAPH:



The plot demonstrates a direct, positive linear relationship between Data Packet Size (Bytes) (X-axis) and Throughput (bps) (Y-axis). As the packet size increases, the fixed 28-byte header overhead (IPv4 + UDP layers) becomes negligible. This allows a much higher percentage of channel bandwidth to be utilized for actual payload data, resulting in higher effective throughput.